Plug & Roll main board. Each block of the board is one sub-sheet.

Input jacks and switching

File: input-jacks.kicad_sch

Input receivers

File: input.kicad_sch

INST buffer

File: input-inst.kicad_sch

Input gain and ADC pad

File: input-gain.kicad_sch

Sheet: /
File: main-board.kicad_sch

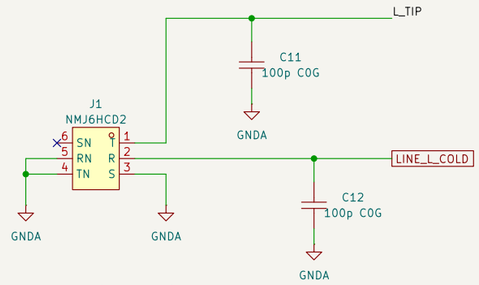
Title: Plug & Roll - Main board

Size: A4	Date:	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 1/5

INPUT JACKS

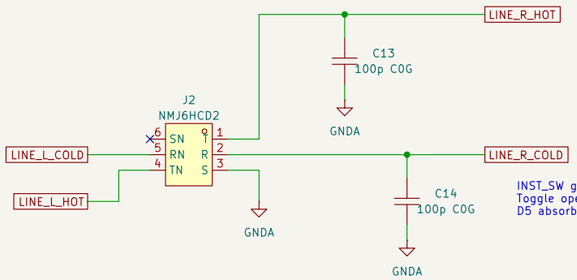
Tip = hot (+), ring = cold (−), sleeve = analog ground.
TN / RN / SN are switch contacts that rest on T / R / S while the jack is empty.

LEFT input



Empty jack: tip and ring are grounded (silent input).

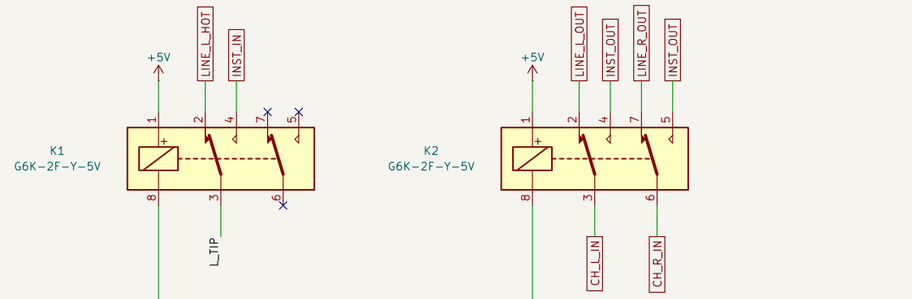
RIGHT input



Empty jack: the Right receiver gets the Left signal (mono).
Purely analog, the Daisy is not involved.

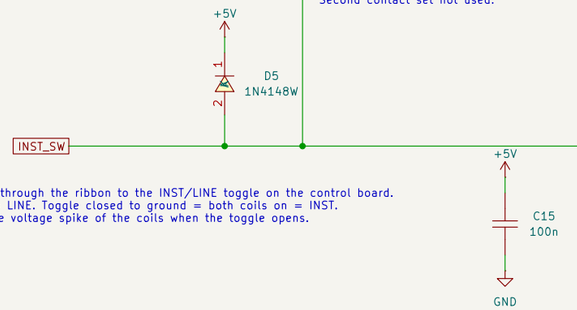
INST / LINE SWITCHING

Relays drawn released (no coil current) = LINE. Panel toggle closed = coils on = INST.



K1: Left tip to the LINE receiver or to the INST buffer.
Second contact set not used.

K2: each channel takes its LINE receiver or the INST buffer.

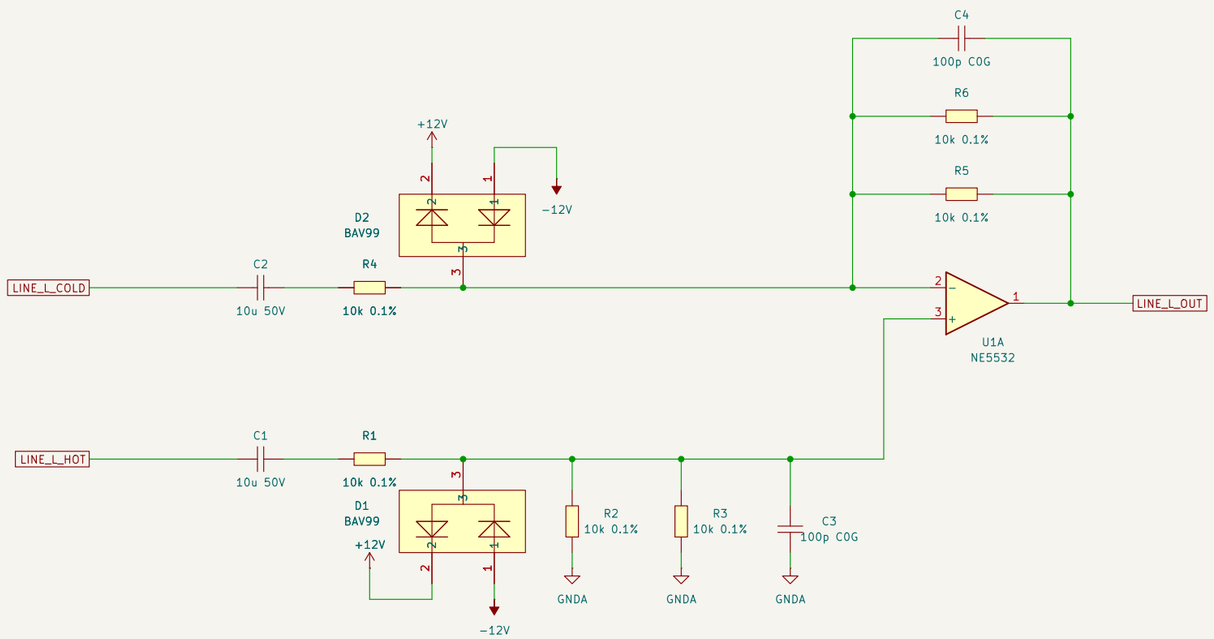


INST_SW goes through the ribbon to the INST/LINE toggle on the control board.
Toggle open = LINE. Toggle closed to ground = both coils on = INST.
D5 absorbs the voltage spike of the coils when the toggle opens.

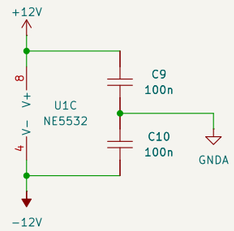
5 V decoupling, placed at the relays

Sheet: /Input jacks and switching/		
File: input−jacks.kicad_sch		
Title: Plug & Roll − Main board − Input jacks and INST/LINE switching		
Size: A3	Date:	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 2/5

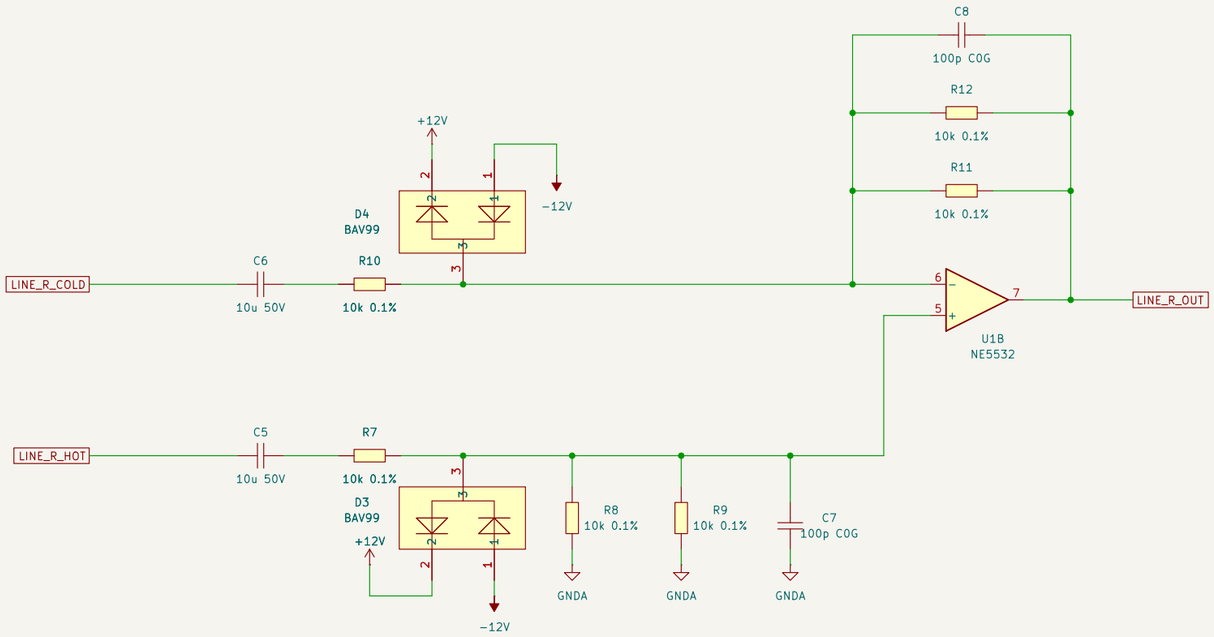
LINE receiver L: balanced input, gain x0.5 (−6 dB). All resistors 10k 0.1 %.



U1 supply and decoupling



LINE receiver R: balanced input, gain x0.5 (−6 dB). All resistors 10k 0.1 %.



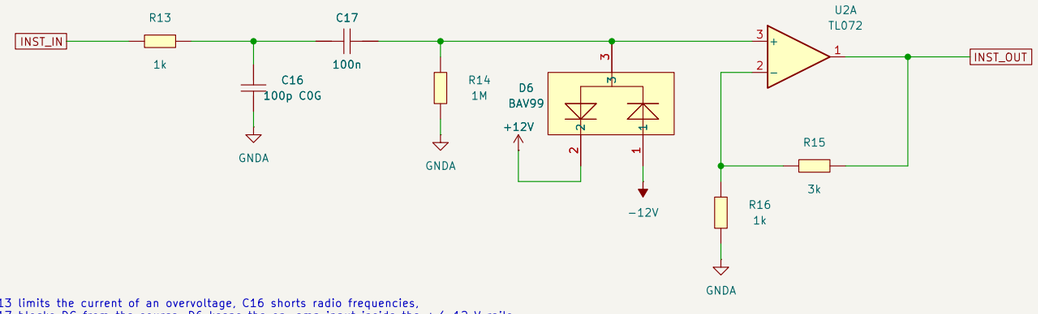
Sheet: /Input receivers/
File: input.kicad_sch

Title: Plug & Roll – Main board – Input receivers

Size: A3	Date:	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 3/5

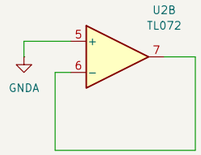
INST BUFFER

Instrument (guitar) input, used only in INST mode. Input resistance 1 Mohm, gain x4 (+12 dB) = 1 + R15/R16.

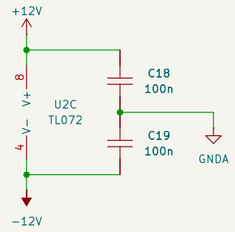


R13 limits the current of an overvoltage, C16 shorts radio frequencies,
C17 blocks DC from the source, D6 keeps the op-amp input inside the +/−12 V rails.

Unused half of U2



U2 supply and decoupling



Sheet: /INST buffer/
File: input-inst.kicad_sch

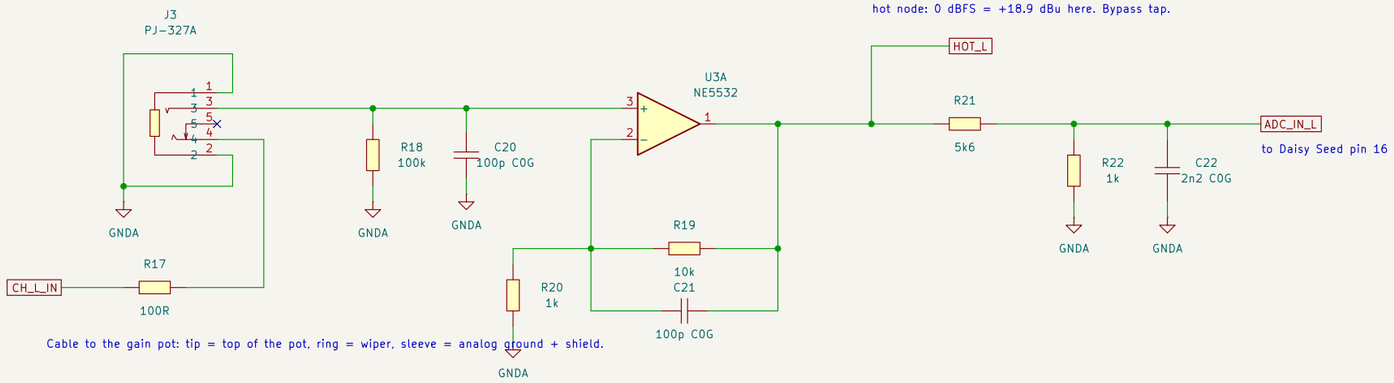
Title: Plug & Roll – Main board – INST buffer

Size: A3	Date:	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 4/5

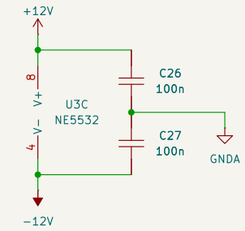
INPUT GAIN AND ADC PAD

The gain pot (10k dual, audio taper) is on the control board. One shielded 3.5 mm stereo cable per channel connects it to J3 / J4.
Pad 5k6 / 1k brings the hot node down to codec level and limits voltage and current at the Daisy pin. No bias: the Seed has its own input capacitor.

LEFT channel: gain pot, x11 (+20.8 dB), pad to the Daisy



U3 supply and decoupling



RIGHT channel: gain pot, x11 (+20.8 dB), pad to the Daisy

Sheet: /Input gain and ADC pad/		
File: input-gain.kicad_sch		
Title: Plug & Roll — Main board — Input gain and ADC pad		
Size: A3	Date:	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 5/5